Output Form (OF)

Score: 4/5 — Minor gaps | Confidence: high

Benchmark: *MASSIVE* | ***Context:*** *Ethiopian Intercity Travelers — Amharic-Language Booking Bot*

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

MASSIVE's output format — intent classification + token-level slot-fill sequence [Q70, Q71, Q73] — directly matches what an intent-and-slot booking system requires. Three primary metrics are reported: 'exact match accuracy, intent classification accuracy, and slot-filling F1 score' [Q5], with per-language breakdowns [Q126, Q127, Q128] and 95% confidence intervals [Q89]. Per-language Amharic results are reported, allowing direct estimation of expected performance. However, two structural concerns remain: (1) Pearson correlation of 0.54 between XLM-R pretraining data quantity and zero-shot exact match [Q87], combined with the finding that 'Germanic genera and Latin scripts performed the best overall' [Q98] and that low-pretraining-data languages like Icelandic underperformed [Q100], strongly suggests Amharic — a unique-script, lower-resource language — will fall in the lower-performance tier; (2) deployment may eventually require text-to-speech for accessibility (literacy ~51.8% in 2017, urban–rural gap [WEB-3, WEB-12, WEB-11]), but MASSIVE evaluates only text outputs [Q5]. OF was rated LOWER priority and overall format alignment is strong.

ID	Evidence
Q5	“We also present modeling results on XLM-R and mT5, including exact match accuracy, intent classification accuracy, and slot-filling F1 score.” (p.1)
Q70	“The first classification head used the pooled output from the encoder to predict the intent and the second used the sequence output to predict the slots.” (p.7)
Q71	“As pooling for the intent classification head, we experimented with using hidden states from the first position, averaged hidden states across the sequence, and the maximally large hidden state from the sequence.” (p.7)
Q73	“The decoder output is a sequence of labels (including the Other label) for all of the tokens followed by the intent.” (p.7)
Q87	“We compared the pretraining data quantities by language for XLM-R to its per-language task performance values, and in the zero shot setup, we found a Pearson correlation of 0.54 for exact match” (p.7)
Q89	“Intervals for 95% confidence are given assuming normal distributions.” (p.8)
Q98	“Discounting for artificial spacing effects, Germanic genera and Latin scripts performed the best overall (See Appendix E), which is unsurprising given the amount of pretraining data for those genera and scripts, as well as the quantity of Germanic and Latin-script languages in MASSIVE.” (p.8)
Q100	“Icelandic was the lowest-performing Germanic language, likely due to a lack of pretraining data, as well as to its linguistic evolution away from the others due to isolated conditions.” (p.8)
Q126	“Exact match accuracy by language for our three models using the full dataset and the zero-shot setup.” (p.20)
Q127	“Table 8: Intent accuracy by language for our three models using the full dataset and the zero-shot setup.” (p.21)
Q128	“Micro-averaged slot-filling F1 by language for our three models using the full dataset and the zero-shot setup.” (p.22)
WEB-3	Ethiopia adult literacy ~51.8% (2017) (https://www.cia.gov/the-world-factbook/about/archives/2023/countries/ethiopia/)
WEB-11	Significant urban–rural literacy gap; pastoralist regions much lower (https://zoetalentsolutions.com/education-statistics-for-ethiopia/)
WEB-12	UNESCO literacy data corroborates ~51.8% figure (https://www.theglobaleconomy.com/Ethiopia/Literacy_rate/)

Strengths

- Output format (intent label + slot-token sequence) directly matches deployment requirement [Q70, Q73]
- Three primary metrics (exact match, intent accuracy, slot F1) match standard NLU evaluation [Q5]

- Per-language results published for Amharic [Q126, Q127, Q128] allowing direct expected-performance estimation
- Reports both full-dataset and zero-shot performance [Q83, Q85] enabling lower-bound estimation for low-resource transfer scenarios

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Q5	“We also present modeling results on XLM-R and mT5, including exact match accuracy, intent classification accuracy, and slot-filling F1 score.” (p.1)
Q70	“The first classification head used the pooled output from the encoder to predict the intent and the second used the sequence output to predict the slots.” (p.7)
Q73	“The decoder output is a sequence of labels (including the Other label) for all of the tokens followed by the intent.” (p.7)
Q83	“Zero-shot performance was also assessed, in which the models were trained on English data, validation was performed on all languages, and testing was performed on all non-English locales.” (p.7)
Q85	“Zero-shot exact match performance is 25-37 points worse than that of full-dataset training runs.” (p.7)
Q126	“Exact match accuracy by language for our three models using the full dataset and the zero-shot setup.” (p.20)
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Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Yes — text-based intent + slot output matches the booking-bot deployment exactly [Q70, Q71, Q73].

OF-2: MASSIVE evaluates text only [Q5]; if accessibility for lower-literacy users requires TTS output (literacy ~51.8% adult, [WEB-3, WEB-12]; rural literacy lower [WEB-11]), MASSIVE provides no signal on TTS quality. INSUFFICIENT DOCUMENTATION on speech-output evaluation.

OF-3: Adult literacy ~51.8% (2017 data, [WEB-3, WEB-12]); the deployment's smartphone-app cohort is a higher-literacy self-selected subset, but accessibility for less literate or older travelers may require speech output not evaluated by MASSIVE.

OF-4: Form mismatches are minor for the text-NLU output but expected Amharic performance is in the lower tier per the script/pretraining-data correlation [Q87, Q98, Q100]; this is a performance-distribution concern rather than a form mismatch.

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Information Gaps

- Reported MASSIVE Amharic exact-match / intent-acc / slot-F1 numbers (Tables 7–9, [Q126, Q127, Q128]) were not surfaced in the documentation extract

Requires Expert Verification

- Whether the deployment ultimately requires TTS output for accessibility, beyond text-only NLU

Remediation

Gap 1: Expected lower-tier performance for Amharic per pretraining-data and script correlations [Q87, Q98, Q100]; reliance on text-only output may underserve lower-literacy users (~51.8% adult literacy)

Recommendation: Report Amharic-specific exact-match, intent-accuracy, and slot-F1 separately rather than aggregated; set deployment thresholds based on Amharic-only metrics. If accessibility for lower-literacy users is in scope, supplement with a TTS-output evaluation track outside MASSIVE.

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